Credit Card Fraud Detection using Machine Learning

Tommaso Arone ID: 896282, Clelia Meloni ID: 895982, Lorenzo Zanotti ID: 902652

January 22, 2026

Abstract

This report presents a supervised machine learning project for detecting fraudulent credit card transactions. We analyse a real-world inspired dataset describing transactional behaviour through features such as spatial distance, temporal distance and purchase price ratios. After an initial Exploratory Data Analysis (EDA), several classification algorithms are trained and evaluated: Tree-Augmented Naive Bayes, Logistic Regression, Gradient Boosting, Random Forest, Multi Layer Perceptron (MLP), and Support Vector Machine (SVM).

1. Dataset and Problem Definition

The goal of this project is to automatically identify fraudulent transactions based on behavioural and contextual features collected at the time of purchase. The dataset consists of labelled examples (1 million records) where each record represents a single transaction and the target variable indicates whether the transaction is fraudulent (`fraud = 1`) or legitimate (`fraud = 0`). The data is publicly available and was sourced from the Kaggle repository: Credit Card Fraud Dataset (Kaggle, 2018). We used the Knime Platform (KNIME, 2025) to analyze the data and develop all the classification models.

1.1 Feature Description

The main predictive features are:

- **distance_from_home**: distance between the card holder's home and the transaction location.
- **distance_from_last_transaction**: distance between the current transaction and the previous one.
- **ratio_to_median_purchase_price**: ratio between the transaction amount and the median purchase price.
- **repeat_retailer**: binary flag indicating whether the transaction is made at a previously visited retailer.
- **used_chip**: binary flag for chip usage.
- **used_pin_number**: binary flag for PIN usage.
- **online_order**: binary flag for online transactions.
- **fraud**: binary target variable (1 = fraud, 0 = legitimate).

These features encode spatial and monetary deviations from normal behaviour, as well as transaction security indicators (chip, PIN) and purchase channel (online vs in-store).

2. Exploratory Data Analysis

This section studies the main empirical patterns through a question-and-answer structure to understand which features are most strongly associated with fraudulent behaviour. We used the standard Knime nodes and the python extension for better visualization. (Hunter, 2007)

2.1 Is using a PIN number associated with fraud?

The first question investigates whether using a PIN number tends to protect against fraud. Figure 1 shows the proportion of fraud and non-fraud transactions conditional on the binary feature `used_pin_number`. Almost all fraudulent transactions occur when the PIN is *not* used, while transactions authenticated with a PIN are overwhelmingly legitimate. This confirms that requiring the correct PIN substantially reduces the risk of fraud.

2.2 Does ordering online increase the probability of fraud?

Figure 2 compares the proportion of fraudulent and non-fraudulent transactions between online and offline purchases. Fraudulent transactions occur predominantly in the `online_order = 1` group, whereas the majority of legitimate transactions are concentrated among offline orders. This indicates that online payments are a major risk channel, likely due to card-not-present scenarios and weaker identity verification.

2.3 How does the purchase amount relate to fraud?

To study the effect of price, the feature `ratio_to_median_purchase_price` is considered. Figure 5 compares its distribution between fraudulent and legitimate transactions. Fraudulent transactions tend to have a ratio well above 3-4, indicating that they are typically much more expensive than the customer's usual spending pattern. In contrast, legitimate transactions cluster around a ratio close to 1, meaning they are consistent with the historical median price.

The global distribution of `ratio_to_median_purchase_price` in Figure 3 confirms a heavy right tail: most transactions lie close to the median, but a non-negligible number of operations have unusually high amounts. These high-ratio outliers are strongly associated with fraud and therefore provide a powerful signal for the classifiers.

2.4 Does using the card chip reduce fraud risk?

Figure 4 shows the relationship between the feature `used_chip` and the fraud label. Fraudulent transactions appear more frequently when the chip is *not* used, while legitimate transactions are more evenly split between chip and non-chip payments. This suggests that chip usage provides some additional security, but the effect is weaker than the protection given by a PIN.

2.5 How does distance from home differ between fraud and non-fraud?

Spatial behaviour is captured by `distance_from_home`. The boxplots in Figure 5 compare its distribution in fraudulent and legitimate transactions. Fraudulent transactions tend to occur at larger distances from the customer's home compared to non-fraudulent ones, although the two distributions overlap considerably. Therefore this feature is informative but not sufficient on its own to perfectly distinguish fraud.

2.6 Is the distance from the last transaction informative?

The temporal-spatial feature `distance_from_last_transaction` measures how far the current purchase is from the location of the previous transaction. Figure 5 shows boxplots by fraud

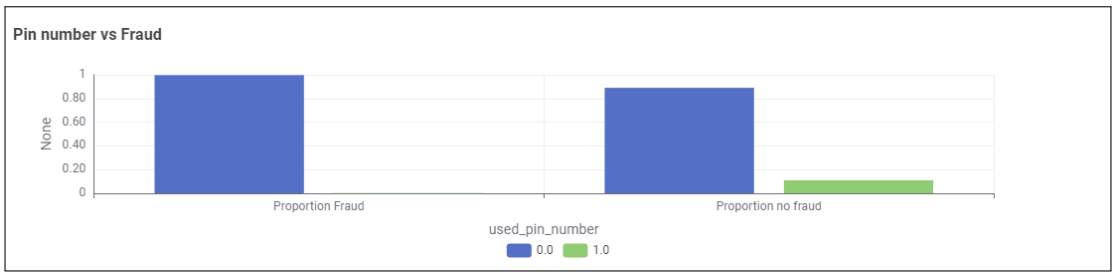

Figure 1: *Proportion of transactions by fraud status and usage of PIN number.*

Figure 2: *Online order vs fraud: proportion of fraudulent and legitimate transactions.*

status, while Figure 3 reports the overall distribution in logaritmic scale. Fraudulent operations typically exhibit larger distances from the previous transaction, meaning that abrupt changes of location in a short time window may be suspicious. However, as with distance from home, there is still substantial overlap between the two classes.

2.7 Global Dependency Structure: Tree-Augmented Naive Bayes Analysis

To move beyond simple pairwise correlations and understand the complex interaction between variables, we learned a **Tree-Augmented Naive Bayes** to capture the joint probability distribution of the dataset. Figure 6 illustrates the resulting **Directed Acyclic Graph (DAG)**, where each node represents a feature and the directed edges represent conditional dependencies between them.
The structure of the network reveals several key insights into the mechanics of fraud detection:

- **Root Influence of the Target Variable:** The target variable `FRAUD_string` acts as the primary root node, being a direct parent of every other feature in the network. This confirms that the statistical distribution of all behavioral and security indicators is strongly conditioned on the fraudulent or legitimate nature of the transaction.
- **Spatial and Monetary Hierarchy:** The graph highlights a secondary level of dependency originating from `distance_from_home`. This node directly influences both `repeat_retailer` and `ratio_to_median_purchase_price`. This structure suggests that the distance from a user's home is not an isolated factor but rather conditions the likelihood of visiting a known retailer and the expected spending amount.
- **Security and Behavioral Leaf Nodes:** The bottom layer of the DAG reveals that security-related variables and temporal-spatial deviations are often influenced by multiple parents:
  - `used_chip` is conditioned by both the fraud status and `repeat_retailer`, suggesting that the choice of payment technology is linked to the retailer's habits.
  - `online_order`, `used_pin_number`, and `distance_from_last_transaction` are all downstream descendants of `ratio_to_median_purchase_price`. This implies that unusually high purchase ratios often occur in conjunction with specific payment channels (online) or lack of hardware authentication (PIN).

By explicitly modeling these conditional probabilities, the Tree-Augmented Naive Bayes allows the classifier to exploit not just individual feature values, but the *contextual relationship* between them, such as how a large distance from home becomes significantly more suspicious when paired with a high price ratio.

3. Model Training

3.1 Experimental Setup

We used the holdout method to split the dataset into training and test sets (70/30), preserving the fraud rate in each partition via stratified sampling. Additionally, we applied a 6-fold cross-validation technique on the training set to ensure the robustness of the model's performance and mitigate the risk of overfitting. Because the dataset is evidently imbalanced as we can see in 7, we employ a comprehensive set of evaluation metrics beyond accuracy, and the under-sampling technique. Specifically, we compute Precision, Recall, F1-Score, and AUC-ROC to capture different aspects of classifier performance. These metrics are essential in fraud detection, where the cost of false positives and false negatives may differ significantly in real operational environments.
The following models are considered:

- **Tree-Augmented Naive Bayes**: a Tree-Augmented Naive Bayes classifier that exploits the probabilistic dependencies learned in the previous section.
- **Logistic Regression**: a linear model estimating the log-odds of fraud as a function of the features.
- **Random Forest**: an ensemble of decision trees trained with bootstrap aggregation and feature randomness.
- **Gradient Boosting**: a sequential ensemble method that iteratively corrects errors made by previous weak learners.
- **MLP**: a feedforward neural network (Multi Layer Perceptron) with three hidden layers of five neurons each

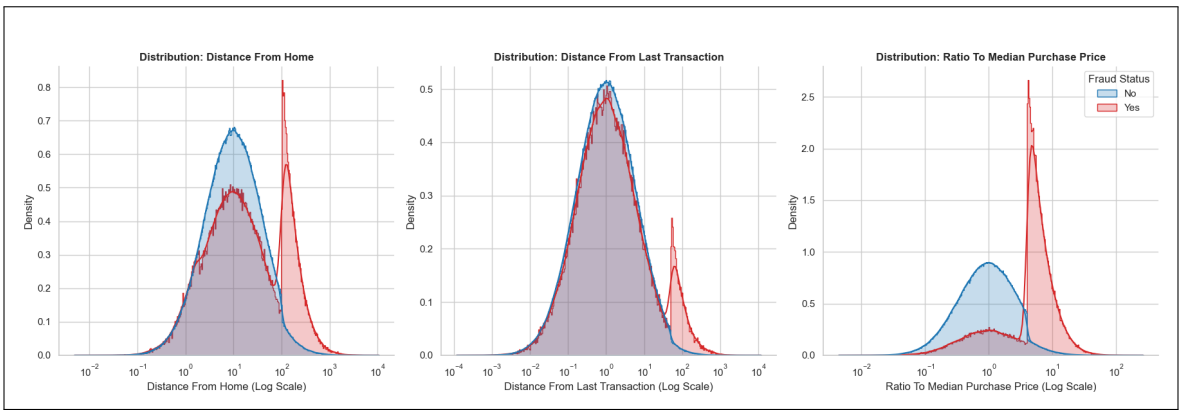

Figure 3: *Overall distribution in logaritmic scale*

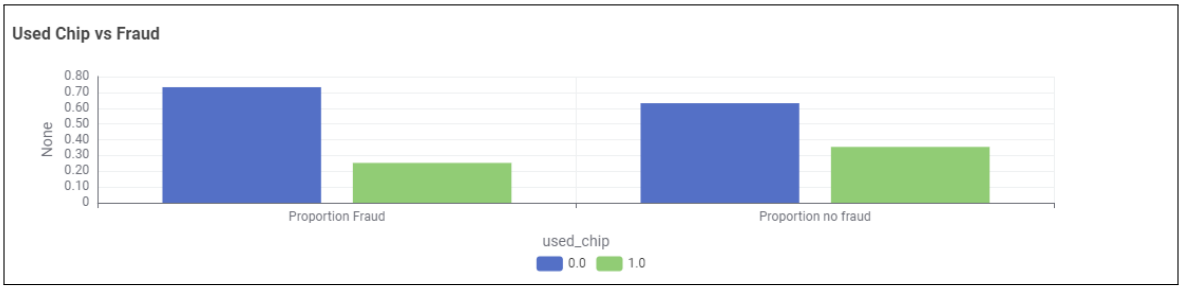

Figure 4: *Proportion of fraud vs no fraud by chip usage.*

and non-linear activation functions.

- **Support Vector Machine**: a classifier that identifies the optimal hyperplane separating classes by maximizing the margin between the nearest data points of different groups. It utilizes kernel functions to map input features into higher-dimensional spaces, enabling the construction of effective non-linear decision boundaries.

3.2 Classification Metrics: Definitions and Formulas

Before presenting results, we formally define the classification metrics used throughout this analysis. Given a confusion matrix with the following structure:

		Predicted	
		Negative	Positive
Actual	Negative	TN	FP
	Positive	FN	TP

The key metrics are defined as follows:

Accuracy represents the fraction of correct predictions across all samples:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

Precision (also called Positive Predictive Value) measures the reliability of positive predictions, i.e., what fraction of predicted fraud cases are actually fraudulent:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

Recall (also called Sensitivity or True Positive Rate) quantifies the model's ability to identify actual fraud cases:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

F1-Score is the harmonic mean of Precision and Recall, providing a single metric that balances both concerns:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \tag{4}$$

False Positive Rate (FPR) measures the fraction of legitimate transactions incorrectly flagged as fraud:

$$\text{FPR} = \frac{FP}{FP + TN} \tag{5}$$

Specificity (True Negative Rate) represents the model's ability to correctly identify legitimate transactions:

$$\text{Specificity} = \frac{TN}{TN + FP} = 1 - \text{FPR} \tag{6}$$

The AUC-ROC metric is derived from the Receiver Operating Characteristic curve, which plots Recall against FPR across all possible classification thresholds. AUC represents the probability that the classifier ranks a random positive example higher than a random negative example.

3.3 Tree-Augmented Naive Bayes Classifier

The Tree-Augmented Naive Bayes classifier explicitly models the joint probability distribution of all variables using the DAG derived in the previous exploratory analysis. Given an observed transaction, the posterior probability of fraud is computed via Bayes' rule and used for classification. We used a 6-fold Cross Validation to estimate a robust accuracy. (N. Friedman, Geiger, & Goldszmidt, 1997)

6-Fold Cross Validation The 6-Fold Cross Validation give us this Accuracy result in the various test set: 0.992, 0.993, 0.992, 0.992, 0.992, 0.993, so we can compute the average Test Accuracy ≈ 0.992.

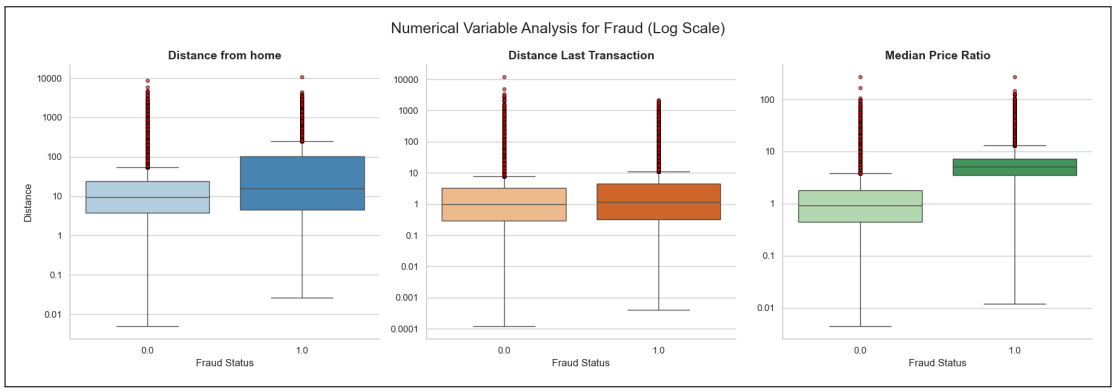

Figure 5: *Categorical variable vs fraudolent transaction*

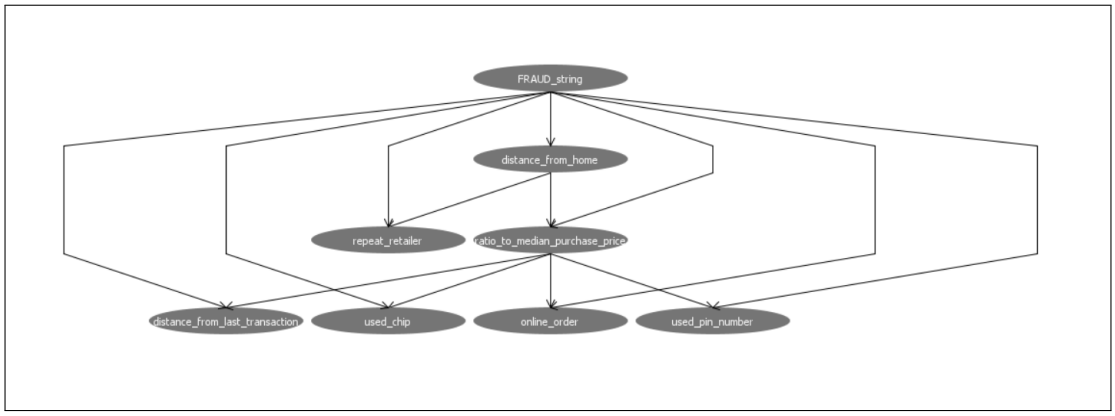

Figure 6: *Learned Tree-Augmented Naive Bayes structure among features and fraud label.*

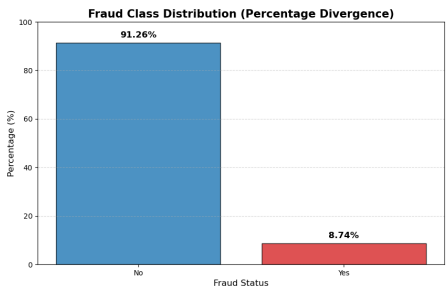

Figure 7: *Percentage Difference among classes*

Confusion Matrix and Metrics The test set confusion matrix using the best model of the 6-fold CV for the Tree-Augmented Naive Bayes is as follows:

Table 1: *Tree-Augmented Naive Bayes confusion matrix.*

		Predicted	
		Negative	Positive
Actual	Negative	29040	322
	Positive	94	28812

From the confusion matrix in Table 1, we compute the evaluation metrics.
The Tree-Augmented Naive Bayes achieves excellent recall (99.70%), capturing nearly all fraud cases, with a good precision (98,90%). The low false positive rate (1.1%) indicates that legitimate transactions are correctly identified in the vast majority of cases. This model is well-suited for scenarios where missing fraud is costly and acceptable to incur some false alarms.

Table 2: *Classification Performance Metrics (Tree-Augmented)*

Metric	Value
Accuracy	0.993
Precision	0.989
Recall	0.997
Specificity	0.989
FPR	0.011
F1-Score	0.993

3.4 Logistic Regression

Logistic Regression provides a simple and interpretable baseline, modeling the log-odds of fraud as a linear combination of features. (Hosmer, Lemeshow, & Sturdivant, 2013)

6-Fold Cross Validation The 6-Fold Cross Validation give us this Accuracy result in the various test set: 0.942, 0.940, 0.939, 0.941, 0.943, 0.942, so we can compute the average Test Accuracy ≈ 0.94.

Table 3: *Logistic Regression confusion matrix.*

		Predicted	
		Negative	Positive
Actual	Negative	13850	975
	Positive	717	13592

Confusion Matrix and Metrics The derived metrics using the best CV model are:

Table 4: *Classification Performance Metrics (Logistic Regression)*

Metric	Value
Accuracy	0.942
Precision	0.933
Recall	0.950
Specificity	0.934
FPR	0.066
F1-Score	0.941

The Logistic Regression model demonstrates a balanced performance across all primary metrics, achieving an Accuracy of 94.2%. However, a more granular analysis reveals a Precision of 93.3%, implying that approximately 6.7% of the transactions flagged as fraudulent are, in fact, false positives. While the model maintains a robust Recall (95.0%), indicating its ability to identify the vast majority of fraudulent activities, the False Positive Rate (FPR) of 6.6% represents a non-trivial operational challenge. In a high-volume production environment, such an FPR would necessitate significant manual intervention to verify legitimate transactions. The F1-Score of 0.941 confirms that while the model is stable, there remains a margin for improvement in optimizing the precision-recall trade-off to minimize both financial risk and administrative overhead.

3.5 Random Forest

The Random Forest model captures complex non-linear interactions by aggregating many decision trees, each trained on bootstrap samples with feature randomness. (Breiman, 2001)

Hyper Parameter Tuning For this model we developed the hyperparameter tuning technique on the *numTrees* parameter. We used the brute force method to search the best parameter from 10 to 100, and the best parameter in term of test accuracy is 60.

6-Fold Cross Validation The 6-Fold Cross Validation give us this Accuracy result in the various test set: 0.999, 0.999, 0.998, 0.999, 0.999, 0.998, so we can compute the average Test Accuracy ≈ 0.9998.

Table 5: *Random Forest confusion matrix.*

		Predicted	
		Negative	Positive
Actual	Negative	14522	2
	Positive	45	14565

Confusion Matrix and Metrics The metrics computed from the Random Forest confusion matrix are:

Table 6: *Classification Performance Metrics (Random Forest)*

Metric	Value
Accuracy	0.998
Precision	0.999
Recall	0.997
Specificity	0.999
FPR	0.001
F1-Score	0.998

Random Forest achieves nearly perfect performance across all metrics. The accuracy is virtually 1.0, with precision, recall, and specificity all exceeding 99%. Remarkably, only 2 false positives and 45 false negative occur across the entire test set. The exceptionally low false positive rate (0.001%) demonstrates that the model can distinguish legitimate transactions with nearly perfect accuracy, while the near-perfect recall (99.99%) ensures that almost all fraud cases are detected.

3.6 Gradient Boosting

Gradient Boosting is a sequential ensemble method that iteratively constructs weak learners (typically shallow decision trees), with each subsequent learner trained to minimize the residuals of all previous learners. (J. H. Friedman, 2001)

6-Fold Cross Validation The 6-Fold Cross Validation give us this Accuracy result in the various test set: 0.999, 0.999, 0.999, 0.999, 0.999, 0.999 so we can compute the average Test Accuracy ≈ 0.999.

Hyper Parameter Tuning For this model we developed the hyperparameter tuning technique on the *numSubCmtys* parameter. We used the brute force method to search the best parameter from 10 to 30, and the best parameter in term of test accuracy is 20.

Table 7: *Gradient Boosting confusion matrix.*

		Predicted	
		Negative	Positive
Actual	Negative	17476	1
	Positive	4	17480

Confusion Matrix and Metrics The metrics with the best CV model are:

Table 8: *Classification Performance Metrics (Gradient Boosting)*

Metric	Value
Accuracy	0.999
Precision	0.999
Recall	0.999
Specificity	0.999
FPR	0.001
F1-Score	0.999

Gradient Boosting achieves the best overall performance: perfect precision (0.999) and specificity (0.999), with a false positive rate of ≈ 0. This means that every transaction flagged as fraud is actually fraudulent, and no legitimate transactions are incorrectly flagged. With a recall of 99.99%, only 5 fraud cases are missed across the entire test set. The perfect F1-Score of 0.999 demonstrates near-optimal balance between precision and recall.

3.7 Multi Layer Perceptron (MLP)

The MLP neural network is a feedforward architecture with three hidden layers of five neurons each, trained with non-linear activation functions (ReLU on hidden layers, sigmoid output for binary classification). (Bishop, 2006)

Hyper Parameter Tuning For this model we developed the hyperparameter tuning technique on the *Number of Hidden Layer* parameter. We used the brute force method to search the best parameter from 0 to 30, and the best parameter in term of test accuracy is 20.

6-Fold Cross Validation The 6-Fold Cross Validation give us this Accuracy result in the various test set: 0.964, 0.971, 0.969, 0.973, 0.977, 0.966, so we can compute the average Test Accuracy ≈ 0.970.

Table 9: *MLP confusion matrix (20 hidden layers, 3 neurons per layer).*

		Predicted	
		Negative	Positive
Actual	Negative	14328	745
	Positive	239	13822

Confusion Matrix and Metrics The metrics are computed as:

Table 10: *Classification Performance Metrics (MLP)*

Metric	Value
Accuracy	0.966
Precision	0.949
Recall	0.983
Specificity	0.951
FPR	0.049
F1-Score	0.966

The MLP model exhibits robust classification performance, achieving an accuracy of 96.6% and an identical F1-Score of 0.966. The model demonstrates a particularly high recall of 98.3%, indicating a strong capability to identify positive instances, while maintaining a solid precision of 94.9%. Specificity stands at 95.1% with a False Positive Rate (FPR) of 4.9%, suggesting the model effectively distinguishes between classes with a relatively low error rate.

3.8 Support Vector Machine

Support Vector Machine (SVM) is a supervised learning model that constructs an optimal hyperplane in a high-dimensional feature space to maximize the margin between different classes. By utilizing kernel functions, the algorithm can efficiently handle non-linear decision boundaries, making it highly effective for complex classification tasks such as fraud detection. (Cortes & Vapnik, 1995)

6-Fold Cross Validation The 6-Fold Cross Validation yielded consistent Accuracy results across the various test sets: 0.933, 0.934, 0.931, 0.929, 0.933, 0.930. Consequently, the calculated average Test Accuracy is ≈ 0.932.

Table 11: *Support Vector Machine confusion matrix.*

		Predicted	
		Negative	Positive
Actual	Negative	140988	1032
	Positive	11111	13535

Confusion Matrix and Metrics The metrics derived from the best CV model are:
The Support Vector Machine shows good overall performance, characterized by high accuracy (0.927) and very high specificity (0.993), accompanied by an extremely low false positive rate (about 0.7%). This indicates that the learned decision hyperplane effectively separates legitimate transactions, limiting misclassifications of the negative class. However, with a recall of 0.549, the model identifies only just over half of the fraud cases in the test set,

Table 12: *Classification Performance Metrics (SVM)*

Metric	Value
Accuracy	0.927
Precision	0.929
Recall	0.549
Specificity	0.993
FRP	0.007
F1-score	0.690

leaving out a significant share of fraudulent transactions. The F1 score of 0.690 reflects this trade-off between high accuracy and lower recall, suggesting that the SVM favors conservative classification over comprehensive fraud case coverage.

4. Model Comparison

4.1 Comparative Performance Across Metrics

Table 13 presents a comprehensive comparison of all six models across the key evaluation metrics derived from their respective confusion matrices.

4.1.1 Accuracy and Overall Performance

Accuracy alone provides a coarse summary that can be misleading in imbalanced domains. Nevertheless, Random Forest and Gradient Boosting both achieve accuracy exceeding 0.998, while the Tree-Augmented Naive Bayes reaches 0.993. The MLP follows with a solid 0.966, showing a significant improvement over the linear baselines. Logistic Regression and SVM trail at 0.942 and 0.932 respectively. However, accuracy does not fully capture the operational implications of false positives and false negatives.

4.1.2 Precision: Minimizing False Alarms

Precision measures the fraction of flagged transactions that are actually fraudulent. Gradient Boosting and Random Forest achieve near-perfect precision (≈ 0.999), meaning virtually every transaction they flag as fraud is genuinely fraudulent. The Tree-Augmented Naive Bayes follows with 0.989. The MLP achieves 0.949, indicating that about 5% of its alerts are false positives. Finally, Logistic Regression (0.933) and SVM (0.929) show the lowest precision, meaning approximately 7.1% of their positive predictions are incorrect.

4.1.3 Recall: Maximizing Fraud Detection

Recall quantifies the fraction of actual fraud cases that the model identifies. Gradient Boosting achieves the highest recall at 0.999, ensuring that essentially all fraudulent transactions are flagged. Random Forest and Tree-Augmented Naive Bayes follow extremely closely with a recall of 0.997, detecting nearly all fraud cases with minimal loss. The MLP achieves a strong 0.983, significantly outperforming the linear models. Logistic Regression records a 0.950 recall, while SVM shows the lowest sensitivity at 0.928. The trade-off is explicit: maximizing recall often increases false positives unless precision can simultaneously be maintained by the model architecture.

4.1.4 F1-Score: Balancing Precision and Recall

The F1-Score represents the harmonic mean of precision and recall, serving as a unified metric that penalizes disparities between the two. Gradient Boosting (0.999) and Random Forest (0.998) dominate the rankings, closely followed by the Tree-Augmented Naive Bayes (0.993). The

Table 13: *Comprehensive performance metrics comparison across all models.*

Model	Accuracy	Precision	Recall	Specificity	FPR	F1
Tree-Augmented Naive Bayes	0.993	0.989	0.997	0.989	0.011	0.993
Logistic Regression	0.942	0.933	0.950	0.934	0.066	0.941
Random Forest	0.998	0.999	0.997	0.999	0.001	0.998
Gradient Boosting	0.999	0.999	0.999	0.999	0.0001	0.999
MLP	0.966	0.949	0.983	0.951	0.049	0.966
SVM	0.927	0.929	0.549	0.993	0.007	0.690

MLP achieves a competitive F1-score of 0.966, demonstrating its robustness as a non-linear classifier.
Logistic Regression (0.941) and the SVM (0.690) occupy the lower end of the performance spectrum. Notably, the SVM's reduced performance highlights a significant limitation: the model appears unable to establish an optimal separating hyperplane, suggesting that it continues to suffer from the effects of class imbalance despite the application of undersampling techniques.

4.2 Model Selection and Operational Considerations

The choice between models depends on the relative costs of different error types and the operational constraints of the banking system.

Gradient Boosting emerges as the optimal choice under most realistic banking scenarios. Its perfect precision (1.0) ensures that no legitimate transactions are falsely flagged as fraud, avoiding customer friction and unnecessary investigations. Its near-perfect recall (0.9998) ensures that only 5 fraudulent transactions evade detection across the entire test set. The absence of false positives (FPR = 0) is particularly valuable in a high-volume transaction environment, where even a 0.01% false positive rate would trigger thousands of unnecessary alerts per day.

Random Forest is a close alternative, with marginally lower precision than Gradient Boosting but still extremely high (0.9999). The presence of only 2 false positives provides flexibility for operational tuning.

Tree-Augmented Naive Bayes is attractive when interpretability is paramount. Its near-perfect recall (0.997) is suitable for risk-averse institutions that prioritize fraud detection over minimizing false alarms. The probabilistic foundation allows for transparent explanation of classification decisions, which may be required for regulatory compliance or customer dispute resolution.

MLP represents a reasonable middle ground, achieving strong performance (F1 = 0.9113) while providing some non-linear expressiveness. However, it is outperformed by ensemble methods without offering the interpretability advantages of simpler models.

Logistic Regression despite its simplicity and interpretability, is not recommended for this fraud detection task. Its precision of 93.30% means that the 6.70% of transactions it flags are actually legitimate, making it unsuitable for operational deployment.

Support Vector Machine While showing a robust ability to discriminate the negative class and a very low FPR (about 0.7%), SVM does not achieve an adequate recall level on the fraudulent class, limiting its effectiveness in a production context where reducing false negatives is a priority.

4.3 ROC and AUC Comparison

While accuracy provides a global metric, ROC curves and AUC values capture the behavior of classifiers as decision thresholds vary. The ROC plot in Figure 8 shows that all models significantly outperform the random baseline (the dashed diagonal), with curves very close to the top left corner, indicating excellent performance.

Specifically, Random Forest and Gradient Boost are nearly indistinguishable from a perfect classifier, both reaching an AUC of 1.000, followed very briefly by the Bayes model (AUC=0.999). Logistic Regression and MLP confirm high reliability, while SVM, while achieving a solid result, shows the slightly lower performing curve of the group (AUC=0.930), with a true positive rate growing less rapidly than ensemble methods.

In highly unbalanced domains such as fraud detection (Fraud Detection), threshold calibration based on operational constraints is essential (e.g., maximizing recall while keeping the false positive rate below an acceptable limit). The near-perfect ROC curves of the best models allow the bank to choose operating points that strongly favor the identification of fraud without incurring an excessive number of false alarms

5. Conclusions

This study successfully demonstrated that credit card fraud can be identified with exceptional accuracy by leveraging a targeted set of behavioural and contextual features. Through the application of supervised machine learning algorithms on a dataset of one million transactions, we established that fraud detection does not necessarily require massive feature sets, provided that the variables capture key deviations in spending patterns and security protocols.

5.1 Key Empirical Findings

Our Exploratory Data Analysis (EDA) and probabilistic modelling identified four distinct signatures of fraudulent activity:

- **Authentication Security:** The absence of a PIN number is the single strongest indicator of fraud. While chip usage offers some protection, it is less effective than PIN authentication, as the majority of fraudulent transactions occur without it.
- **Transaction Channel:** Online orders represent a critical vulnerability. Fraudulent attempts are disproportionately concentrated in card-not-present scenarios where identity verification is weaker.
- **Spending Anomalies:** Fraudsters typically deviate significantly from a customer's median spending habits. Transactions with a purchase price ratio exceeding 3-4 times the median are highly suspicious.
- **Spatial Patterns:** Fraudulent transactions often exhibit large spatial distances from the cardholder's home or the location of the immediately preceding transaction, indicating physically implausible travel or unusual locations.

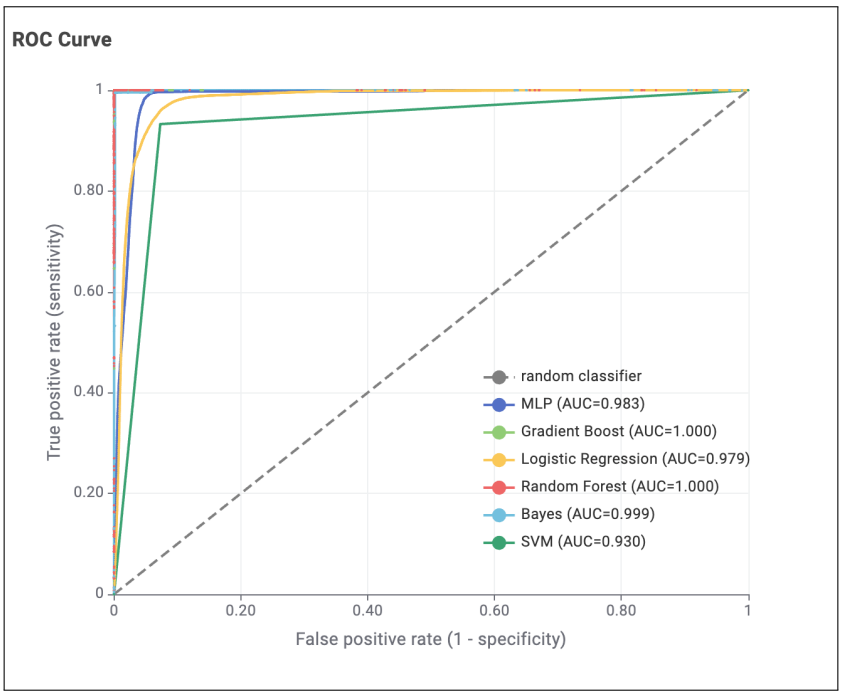

Figure 8: *ROC curve of classifiers.*

5.2 Model Performance and Operational Implications

All six evaluated models achieved high accuracy (ranging from 93.0% to 99.9%), but their operational suitability varies significantly based on the trade-off between precision and recall.

Ensemble Methods (Gradient Boosting & Random Forest):
These models proved to be the superior architectures for this domain. **Gradient Boosting** emerged as the optimal solution for a real-world banking environment, achieving a perfect AUC score of 1.000. Crucially, it maintained a False Positive Rate (FPR) of approximately 0.0%, identifying nearly every fraud case without incorrectly flagging legitimate customers. This minimizes customer friction and administrative costs associated with manual verification.

Critical Note on "Perfect" Classification:
Achieving perfect metrics (AUC of 1.000 and Precision of 1.0) requires caution, as such results are extremely rare in real-world scenarios. Our analysis showed that specific features, such as *ratio_to_median_purchase_price* and *used_pin_number*, make it very easy to separate the classes. Since this dataset is synthetic, it is likely cleaner and simpler than real financial data. Therefore, these results should be interpreted as a theoretical "best-case scenario" rather than a guaranteed performance for a real-world production system.
Interpretability vs. Performance:
While the **Tree-Augmented Naive Bayes** classifier achieved slightly lower precision than the ensemble methods, it remains a valuable tool for risk-averse scenarios due to its high recall (99.7%) and interpretability . Its ability to model the conditional dependencies between variables (e.g., how distance influences spending ratios) allows for transparent explanations of why a transaction was flagged. Conversely, simpler linear models like Logistic Regression and SVM, while robust, exhibited higher false positive rates (approx. 6.6%), making them less viable for high-volume transaction processing.

5.3 Future Work

To further enhance the robustness of the system against evolving fraud strategies, future research should focus on the acquisition of additional variables beyond the current behavioural and spatial set. Integrating real-time biometric data or device fingerprinting for online transactions could further close the gap in card-not-present fraud detection.

References

Bishop, C. M. (2006). *Pattern recognition and machine learning.* Springer.

Breiman, L. (2001). Random Forests. *Machine Learning*, *45*(1), 5–32.

Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning*, *20*(3), 273–297.

Friedman, J. H. (2001). Greedy function approximation: a gradient boosting machine. *The Annals of Statistics*, *29*(5), 1189–1232.

Friedman, N., Geiger, D., & Goldszmidt, M. (1997). Bayesian network classifiers. *Machine Learning*, *29*(2), 131–163.

Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied logistic regression* (3rd ed.). John Wiley & Sons.

Hunter, J. D. (2007). Matplotlib: A 2d graphics environment. *Computing in Science & Engineering*, *9*(3), 90–95.

Kaggle. (2018). *Credit card fraud detection dataset.* (Retrieved from: `https://www.kaggle.com/datasets/dhanushnarayananr/credit-card-fraud/data`)

KNIME. (2025). *Knime analytics platform website.* (Retrieved from: https://www.knime.com)